Thomas Farinelli

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SUMMARY

My academic work has focused on Machine Learning, Deep Learning, biomedical data analysis, medical image processing, eye-tracking data, statistical analysis, and the development of data-processing pipelines. I am particularly interested in the use of Artificial Intelligence as a tool to support medical diagnosis, clinical decision-making, biomarker identification, and the analysis of complex biomedical data.

EDUCATION

Roma Tre University, Rome, Italy Rome, Italy

Master's Degree in Computer Engineering (Specialization: Artificial Intelligence and Machine Learning), Current average grade: 29.5/30, Credits earned: 94/120 CFU, Master's Thesis: Analysis of the Correlation Between Visual Impairments and Eye Tracking Data, Supervisor: Prof. Giuseppe Sansonetti, Co-supervisor: Prof. Marco Barbieri, The thesis investigates the relationship between visual difficulties associated with presbyopia and eye-tracking data collected during reading tasks., Using Tobii Pro Glasses 3, eye movements were recorded while participants read texts with progressively decreasing font sizes., The work focuses on the analysis of fixations, saccades, and regressions, with the objective of identifying changes in oculomotor behavior associated with increasing visual difficulty., A dedicated data-processing pipeline was developed to filter raw eye-tracking data, identify reading sequences, extract relevant ocular events, and perform descriptive and inferential statistical analyses., Presbyopic participants were subsequently compared with a normative control group to identify measurable deviations in oculomotor behavior., The work combines experimental data processing, statistical analysis, biomedical applications, and computational methodologies. *Expected graduation: October 2026*

Roma Tre University, Rome, Italy Rome, Italy

Bachelor's Degree in Computer Engineering (Degree Class: L-8 - Information Engineering), Final grade: 103/110, Credits: 180 CFU, Bachelor's Thesis: Machine Learning Techniques for the Diagnosis of Pulmonary Diseases, The thesis explored the use of Machine Learning techniques for the analysis and diagnosis of pulmonary diseases, with particular attention to the potential of data-driven approaches as tools supporting medical diagnosis and clinical decision-making., This project represented my first academic research experience at the intersection of Artificial Intelligence and medicine and motivated my subsequent interest in biomedical applications of Machine Learning and data analysis. *Graduation date: 21 March 2024*

PROJECTS

Fine-Tuning Small Language Models for Structured Information Extraction | Deep Learning Project

- Academic project focused on fine-tuning Small Language Models to transform unstructured natural-language information into rigid machine-readable JSON representations.
- The experiments were based on a subset of the RecipeNLG dataset.
- Models evaluated: Phi-3 (3.5B), Mistral (7B), Qwen 2.5 (1.5B).
- The models were evaluated before and after fine-tuning.
- LoRA was used for Phi-3 and Qwen 2.5, while QLoRA was used for Mistral.
- LLaMA 3.3 70B was used as an inference-only large-language-model baseline.
- Evaluation considered syntactically valid JSON generation, schema compliance, semantic correctness, and generalization to unseen examples.
- The experiments showed substantial improvements after fine-tuning, particularly for Qwen 2.5.

Systematic Study of Preprocessing Techniques for Histopathological Images Using ResNet-18 | Medical Imaging and Deep Learning Project

- Team members: Thomas Farinelli, Giacomo Cui, Alessandro Di Mario.
- Academic research project investigating the impact of image preprocessing on Deep Learning classification of histopathological images of lung and colon tissues.
- Dataset: approximately 25,000 histopathological images, including benign tissue, adenocarcinoma, and squamous cell carcinoma.
- Seven preprocessing strategies were evaluated: Standard preprocessing, Data augmentation, CLAHE, Wavelet transforms, Retinex and Sobel filtering, Homomorphic and high-pass filtering, Fuzzy enhancement.
- Two ResNet-18 configurations were compared: ImageNet pre-trained ResNet-18 and ResNet-18 trained from scratch.
- Personal contribution: implementation and evaluation of Wavelet preprocessing, Retinex and Sobel filtering, and CLAHE, together with experimentation using pre-trained neural networks.
- Performance evaluated via accuracy, precision, recall, F1-score, confusion matrices, t-SNE, training behavior, false-negative analysis, and Grad-CAM explainability.
- Wavelet preprocessing with pre-trained ResNet-18 achieved approximately 99.72% accuracy with zero false negatives.

Oculomotor Movement Analysis | Research code developed in connection with the Master's thesis for processing and analysing eye-tracking recordings.

- Includes procedures for processing gaze data and extracting relevant oculomotor events.
- Focus on fixations, saccades, regressions, reading behavior, variations associated with font size and visual difficulty, and comparison between normative and presbyopic participants.
- Measures are used in descriptive and statistical analyses to investigate patterns associated with visual impairment.

GitHub Portfolio | The repository contains academic projects and implementations developed during my Computer Engineering studies, particularly in Artificial Intelligence, Deep Learning, Machine Learning, biomedical applications, and data analysis.

TECHNICAL SKILLS

Artificial Intelligence

Machine Learning

Deep Learning

Health Data Science

Biomedical Data Analysis

Artificial Intelligence for Medicine and Healthcare

Medical Image Analysis

Explainable Artificial Intelligence

Eye Tracking and Oculomotor Data Analysis

Statistical Analysis of Biomedical Data

Large and Small Language Models

Clinical Decision Support Systems

Data Processing and Computational Pipelines

Programming and Scientific Computing: Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Jupyter, Git, GitHub.

Artificial Intelligence and Machine Learning: Machine Learning, Deep Learning, supervised and unsupervised learning, neural networks, convolutional neural networks, transfer learning, model fine-tuning, Small Language Models, Large Language Models, LoRA, QLoRA, structured information extraction, model evaluation.

Biomedical AI and Medical Data Analysis: medical image classification, histopathological image analysis, eye-tracking data analysis, oculomotor event analysis, biomedical data processing, clinical decision-support applications.

Data Analysis and Statistics: data preprocessing, descriptive and inferential statistics, non-parametric statistical tests, correlation analysis, dimensionality reduction, t-SNE, experimental data analysis, computational pipeline development.

Explainable AI and Image Processing: Grad-CAM, CLAHE, Wavelet transforms, Retinex and Sobel filtering, image augmentation and preprocessing.

Independent research

Scientific literature review

Experimental design

Critical thinking

Problem solving

Quantitative data analysis

Interpretation of experimental results

Scientific and technical writing

Development of reproducible computational workflows

Teamwork in interdisciplinary projects

Independent and collaborative working

Adaptability in international environments

MASTER'S DEGREE COURSEWORK

Artificial Intelligence — 29/30 - 9 CFU; Software Systems Architecture — 28/30 - 9 CFU; Automata, Languages and Computing — 19/30 - 9 CFU; Machine Learning — 30/30 - 9 CFU; Artificial Intelligence from Engineering to Arts — 30/30 - 6 CFU; Technologies and Architectures for Data Management — 30/30 with honours - 9 CFU; Computer Graphics — 30/30 - 6 CFU; Deep Learning — 30/30 with honours - 6 CFU; Decision Support Systems and Analytics — 25/30 - 6 CFU; Algorithms for Big Data — 28/30 - 6 CFU; Intelligent Systems for the Internet — 30/30 with honours - 6 CFU; Automated Planning — 30/30 with honours - 6 CFU; Data Law — 30/30 - 6 CFU; Skills Useful for Entering the Labour Market — Pass - 1 CFU; Final Examination / Master's Thesis — 26 CFU - in progress.

BACHELOR'S DEGREE COURSEWORK

Mathematical Analysis I — 18/30 - 12 CFU; Geometry and Combinatorics — 30/30 - 12 CFU; Fundamentals of Computer Science — 25/30 - 12 CFU; Chemistry — 24/30 - 6 CFU; Algorithms and Data Structures — 30/30 with honours - 9 CFU; Discrete Event Systems Analysis — 24/30 - 6 CFU; Electrical Engineering and Electronics — 28/30 - 9 CFU; Computer Architecture — 23/30 - 6 CFU; Operations Research I — 24/30 - 6 CFU; Fundamentals of Automatic Control — 27/30 - 9 CFU; Functional Programming — 26/30 - 6 CFU; Database Systems I — 25/30 - 6 CFU; Computer Networks — 22/30 - 6 CFU; Object-Oriented Programming — 22/30 - 9 CFU; Fundamentals of Telecommunications — 22/30 - 9 CFU; Mobile Computing — 30/30 with honours - 6 CFU; Engineering Economics — 29/30 - 6 CFU; Web Information Systems — 22/30 - 6 CFU; Artificial Intelligence and Machine Learning — 27/30 - 6 CFU; Physics I — 24/30 - 12 CFU; Software Analysis and Design — 27/30 - 6 CFU; Internship — Pass - 9 CFU; Final Examination — Pass - 3 CFU; Bachelor's weighted average: 25.11/30.

INTERNATIONAL EXPERIENCE

London, United Kingdom (2017-2019). Living in London provided significant experience in an international and multicultural environment and contributed to strengthening my independence, adaptability, communication skills, and ability to interact with people from different cultural and professional backgrounds.

LANGUAGES

Italian: Native. English: B1 university language qualification.

REFERENCES

Academic references available upon request. Prof. Giuseppe Sansonetti — Roma Tre University, Department of Civil, Computer Science and Aeronautical Technologies Engineering, Email: giuseppe.sansonetti@uniroma3.it. Prof. Fabio

Gasparetti — Roma Tre University, Department of Civil, Computer Science and Aeronautical Technologies Engineering, Email: fabio.gasparetti@uniroma3.it.